

sol1_enhancer_codex

Same loop as Lab 1. Isolation is a **process sandbox**, not a role flag.

Codex has no per-agent tool list. `$enhancer-judge` inside the orchestrator session inherits `workspace-write`. The jail is `bin/role.sh` starting a child `codex exec -s read-only`.

Work from `solutions/sol1_enhancer_codex`.

What you will build

A Codex skill pack that grooms draft tickets the same way Saturday's Claude plugin does.

PIECE	PATH
Orchestrator	<code>.agents/skills/enhancer-loop/SKILL.md</code>
Judge skill	<code>.agents/skills/enhancer-judge/SKILL.md</code>
Doer skill	<code>.agents/skills/enhancer-doer/SKILL.md</code>
Field check	<code>.agents/skills/enhancer-loop/scripts/check_fields.py</code>
Stop check	<code>.agents/skills/enhancer-loop/scripts/check_stop.py</code>
Jail	<code>bin/role.sh</code>
Fence probe	<code>bin/fence_check.sh</code>

`check_fields.py` and `check_stop.py` are byte-identical to the lab.

Why Codex is a different lesson

Claude isolates by omitting Write from YAML. Codex cannot. Every in-session skill inherits the parent's sandbox.

If the orchestrator runs under `workspace-write`, an in-session `$enhancer-judge` can edit the ticket it grades.

The product is still: vague issue in, ready contract out. The new knob is a child process.

Learning objectives

- Configure Codex skills under `.agents/skills/`
- Implement `bin/role.sh` so judge and doer cannot write
- Troubleshoot the four Codex traps
- Validate with `task fence-check` then `task run`
- Deploy the same loop on issue events (`ENHANCER_BACKEND=codex`)

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Starting architecture

```
task run
├─ codex exec -s workspace-write (orchestrator)
│   └─ bin/role.sh enhancer-judge
│       └─ codex exec -s read-only
├─ bin/role.sh enhancer-doer
│   └─ codex exec -s read-only
```

Trigger still lives outside. One poll, then exit. Labels, marker, and three exits are unchanged.

Prerequisites

```
cd solutions/sol1_enhancer_codex
codex --version
cp config.json.example config.json  # fork_owner
task clone
task fence-check
```

`task fence-check` fails if `~/.codex/config.toml` has `trust_level = "trusted"`. That setting silently turns the orchestrator fence off. Judge and doer stay read-only either way.

The fence. Flags live in the script

```
# bin/role.sh. The flags are not in SKILL.md on purpose.
codex exec -s read-only --cd "$DIR" -o "$OUT" "\$$SKILL $*" \
  </dev/null >/dev/null 2>&1
cat "$OUT"
```

< /dev/null is load-bearing. `codex exec` appends open stdin and waits for EOF. `task` pipes stdin. Without the redirect the child hangs with no error.

`--cd` lands in this folder so Codex finds `.agents/skills`. Ticket paths are passed absolute. A read-only process may still read outside its workspace.

task run

```
codex exec
-s workspace-write
-c sandbox_workspace_write.network_access=true
--add-dir {{.TARGET}}
--add-dir {{.CODEX_HOME_DIR}}
"\$enhancer-loop --repo {{.TARGET}} {{.CLI_ARGS}}"
< /dev/null
```

Escape `\$enhancer-loop` or the shell eats `$enhancer`.

`--add-dir $HOME/.codex` is required. A child `codex exec` cannot start without it.

AGENTS.md split by role

A first draft said "run judge through `bin/role.sh`". The judge read it, obeyed, started another judge. Infinite recursion.

Now:

- Judge and doer ignore orchestrator instructions
- Orchestrator never invokes `$enhancer-judge` as an in-session skill
- It shells out through `bin/role.sh`

Read `IMPLEMENTATION_NOTES.md` before you "simplify" this.

Three model processes per round

1. Judge scores the current ticket
2. Doer drafts a replacement body
3. Judge scores the candidate

Budget five minutes per poll. `timeout 420 task run --` while developing.

No self-reinvoke. `codex exec` exits when the turn ends. `task poll-forever` is still the seminar scheduler.

Commands and expected result

```
python3 .agents/skills/enhancer-loop/scripts/check_fields.py --demo
python3 .agents/skills/enhancer-loop/scripts/check_stop.py --demo
task create-test-tickets
timeout 420 task run --
```

Expected: T900/T901/T902 grooms, `enhanced` label after a real rewrite, comment with `<!-- enhancer-loop -->`, state file under `.harness/`.

Second poll with no new human comment: no-op. Must not post again.

Troubleshooting

SYMPTOM	CAUSE	FIX
Silent hang	open stdin	< /dev/null on every codex exec
Nested judges	AGENTS.md recursion	split instructions by role
Orchestrator can write as judge	trust_level = trusted	task fence-check
Child cannot start	missing --add-dir ~/.codex	add it on task run
\$enhancer empty	unescaped \$ in Taskfile	\\$enhancer-loop

Validation checklist

- ☐ `task fence-check` is green
- ☐ `--demo` scripts pass
- ☐ Judge process is `-s read-only`
- ☐ One poll rewrites a stub and labels `enhanced`
- ☐ Marker is on the posted comment
- ☐ LGTM plus green rubric sets `ready`

GitHub Actions

Set `ENHANCER_BACKEND=codex` in the copy-me workflow:

```
labs/lab1_enhancer/workflows/enhance-on-issue.yml
```

Hosted runners must have `codex` installed. If they do not, use Agent SDK or Claude Code on Actions. Keep Codex on a laptop.

The trigger starts one poll. This folder still owns the exits. Skip `<!-- enhancer-loop -->`.

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Recap

Isolation in Codex is a process sandbox, not a role flag.

The loop did not change. Ready is still `check_fields.py`. The judge still has no write path.

Takeaways

1. Flags live in `bin/role.sh`, not in the skill.
2. Close stdin.
3. Split AGENTS.md by role or you recurse.
4. Same three exits as Saturday.